

e& Business Network Operations

Network Troubleshooting Guide

This guide provides comprehensive procedures for troubleshooting network issues in the e& Business Network Operations.

1. Network Connectivity Troubleshooting

When investigating connectivity issues, follow these steps:

Step 1: Verify Physical Connections

Check all physical connections including cables, ports, and network interfaces. Ensure all devices are properly connected.

Step 2: Check Network Interface Status

Use the following commands to verify interface status:

```
show interface status
show interface gigabitEthernet 0/0
show ip interface brief
```

Step 3: Verify IP Configuration

Check IP addressing and subnet configuration:

```
show ip address
show running-config interface
show ip route
```

Step 4: Test Connectivity

Perform connectivity tests:

```
ping destination_ip
tracert destination_ip
show arp
```

2. Performance Issues

When investigating performance degradation:

Step 1: Check CPU and Memory Usage

Monitor device resources:

```
show processes cpu
show memory
show system resources
```

Step 2: Analyze Traffic Patterns

Review network traffic:

```
show interface statistics
show traffic summary
show flow statistics
```

Step 3: Check for Congestion

Identify network bottlenecks:

- show queue statistics
- show buffer utilization
- show interface errors

3. Routing Issues

For routing problems, use these procedures:

Step 1: Verify Routing Table

Check routing information:

- show ip route
- show ip route summary
- show ip route destination_ip

Step 2: Check Routing Protocols

Verify routing protocol status:

- show ip ospf neighbor
- show ip bgp summary
- show ip eigrp neighbors

Step 3: Verify Route Redistribution

Check route distribution:

- show ip route redistributed
- show running-config | section router

4. DNS Resolution Issues

When DNS is not working:

Step 1: Test DNS Resolution

Verify DNS functionality:

- nslookup domain_name
- dig domain_name
- show dns statistics

Step 2: Check DNS Configuration

Review DNS settings:

- show running-config | section dns
- show ip name-server

5. VPN Connectivity

For VPN connection problems:

Step 1: Verify VPN Status

Check VPN connections:

- show vpn status
- show vpn sessions
- show ipsec sa

Step 2: Check VPN Configuration

Review VPN settings:

- show running-config | section vpn
- show crypto map

6. Wireless Network Issues

Troubleshooting wireless connectivity:

Step 1: Check Access Point Status

Verify access point connectivity:

- show wireless ap status
- show wireless clients

Step 2: Verify Signal Strength

Check signal quality:

- show wireless signal
- show wireless statistics

7. Common Resolution Commands

Quick reference for common issues:

- Clear ARP cache: clear arp-cache
- Reset interface: shutdown / no shutdown
- Reload device: reload
- Clear counters: clear counters
- Show logs: show log | tail -100

8. Escalation Procedures

If issues cannot be resolved:

- 1. Document all troubleshooting steps taken**
- 2. Collect relevant logs and diagnostics**
- 3. Escalate to Network Operations Manager**
- 4. Contact vendor support if hardware issue**

9. Maintenance Windows

Scheduled maintenance procedures:

- 1. Notify stakeholders 48 hours in advance**
- 2. Create maintenance ticket**
- 3. Perform changes during approved window**
- 4. Verify service restoration**
- 5. Update documentation**

10. Network Monitoring

Key monitoring commands:

- show snmp statistics
- show monitoring status
- show alert configuration
- show performance metrics